

Monomial $f(q) = a \cdot q^4$ Report

Polynomial

$$f(q) = aq^4$$

Naive

$$g_{\text{naive}}(X) = ax^4$$

$$\mathbb{E}[g_{\text{naive}}(X)] = aq^4 + 6aq^2\sigma^2 + 3a\sigma^4$$

$$\text{Var}(g_{\text{naive}}(X)) = 16a^2q^6\sigma^2 + 168a^2q^4\sigma^4 + 384a^2q^2\sigma^6 + 96a^2\sigma^8$$

$$\text{MSE}(g_{\text{naive}}) = 16a^2q^6\sigma^2 + 204a^2q^4\sigma^4 + 420a^2q^2\sigma^6 + 105a^2\sigma^8$$

$$\text{Bias}(g_{\text{naive}}) = 6aq^2\sigma^2 + 3a\sigma^4$$

Unbiased

$$g_{\text{unb}}(X) = 3a\sigma^4 - 6a\sigma^2x^2 + ax^4$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^4$$

$$\text{Var}(g_{\text{unb}}(X)) = 16a^2q^6\sigma^2 + 72a^2q^4\sigma^4 + 96a^2q^2\sigma^6 + 24a^2\sigma^8$$

$$\text{MSE}(g_{\text{unb}}) = 16a^2q^6\sigma^2 + 72a^2q^4\sigma^4 + 96a^2q^2\sigma^6 + 24a^2\sigma^8$$

Comparison

$$\text{mean gap} = -6aq^2\sigma^2 - 3a\sigma^4$$

$$\text{variance gap} = -96a^2q^4\sigma^4 - 288a^2q^2\sigma^6 - 72a^2\sigma^8$$

$$\text{MSE gap} = -132a^2q^4\sigma^4 - 324a^2q^2\sigma^6 - 81a^2\sigma^8$$

$$\text{Bias}(g_{\text{naive}})^2 = 36a^2q^4\sigma^4 + 36a^2q^2\sigma^6 + 9a^2\sigma^8$$